Time-Lag Analysis: Methodology Summary

LIHEAP, CalFresh and Food Bank Distribution · San Diego County · February–October 2023

1 Lag Structure

For the “Generate time-lagged datasets” checkpoint, I built a dataset of LIHEAP enrollment by zip code and month, with three lag columns holding enrollment one, two and three months earlier. I standardized zip codes as five-digit strings and converted Month_Year to a monthly period so calendar-month arithmetic would work. I then built a complete grid of every zip code and every month from January through October 2023, extending three months past the last month of data so each lag has a row to land on, and merged enrollment onto it. Unreported months were left as missing rather than 0. Building the grid first ensures each lag is a true calendar month, not a value pulled from the wrong month when a zip code has a reporting gap.

The add_lags function shifts each month’s enrollment forward by *k* months and merges it back on zip code and month to fill LIHEAP_lag1-LIHEAP_lag3. Rows with no lag values were removed, which dropped January 2023, so the final dataset covers February through October 2023. A spot_check function compared every lag against the raw enrollment from *k* months earlier for zip codes 91901, 91910, 91944, 92059 and 92106 and found zero mismatches. The result was saved as LIHEAP_lags.csv.

2 Edge Cases

Using the validated data window (January–July 2023), I used np.select to give each lag value one of four statuses: **present**; **before_start**, where the lag month precedes the data; **after_end**, where it follows the data; and **zip_gap**, where the month is inside the window but the zip code has no record. These roll up into two row-level flags: Complete_Lags (all three lags present) and Edge (start, end or core). A crosstab of the two confirmed the expected pattern:

Edge	Rows	Complete lags	Explanation
start	173	0	February and March 2023, whose earliest lags fall before January 2023
end	165	0	September and October 2023, whose lags extend past July 2023
core	514	346	The other 168 are missing at least one lag because of zip reporting gaps

I separated these cases because missing values don’t all mean the same thing. Lags outside the data window are structural and can’t be filled, while zip gaps reflect a zip code’s own reporting history and may matter for data quality. Rows were flagged rather than dropped, so later analyses can use only complete-lag rows or every available pair, depending on the method.

3 Policy Flag Integration

I merged Policy_Annotations.csv from Assignment 4 into LIHEAP_lags.csv by month. Policy is countywide, so every zip code in a given month shares the same policy context. Before merging, I created 0/1 indicators on the full January 2018–August 2025 policy timeline, so February 2023 is compared with its true prior month:

- **Change flags:** Policy_Change for each program, plus Any_Policy_Change, marks the month active policies changed.
- **Period flags:** Emergency Allotments ended (April 2023 onward), ABAWD work requirements resumed (July 2023 onward) and the FY2024 LIHEAP funding cut (October 2023).
- **Direction flags:** LIHEAP's expected increase or decrease.

Change flags mark only the month a shift begins, while period flags stay on afterward, separating the moment of a change from the period that follows. The left join used validate="many_to_one", so a duplicated month would raise an error instead of silently duplicating rows; all 852 rows matched. Grouping by month confirmed each flag had exactly one value per month and switched on in the expected months. The result was saved as TimeLag_PolicyAnnotated.csv.

4 Preliminary Correlation Findings

I merged CalFresh dollars and food bank pounds distributed onto the dataset by zip code and month, treating food bank zeros as missing because they mostly reflected gaps in partner reporting. I correlated LIHEAP enrollment from 0–3 months earlier with each month's values using Spearman's ρ, since LIHEAP counts are small integers and CalFresh dollars are skewed.

LIHEAP vs.	Full period	Policy-stable months	Policy-change months
CalFresh ($)	0.64–0.70	0.65–0.74	0.63–0.73
Food bank (lbs)	0.46–0.47	0.42–0.49	0.44–0.51

Spearman ρ across lags 0–3. All correlations significant at $p < 0.001$.

The correlations barely changed from lag 0 to lag 3, so the relationship reflects geography, with high-need zip codes ranking high on all three programs, rather than LIHEAP enrollment predicting later demand. Within-zip correlations, which remove differences in zip size, support this: food bank correlations fell to about zero, and CalFresh correlations turned negative at longer lags (−0.41 at lag 3). This likely reflects the end of CalFresh Emergency Allotments in April 2023, which lowered CalFresh dollars in every zip code at once.

5 Do Policy Periods Affect Correlation Patterns?

Pooled correlations were nearly identical in policy-stable months (March, May, June and August) and policy-change months (February, April, July, September and October), so the geographic relationship held steady through the 2023 policy changes. The one difference was that negative within-zip CalFresh correlations were stronger in policy-change months (−0.47 versus −0.15 at lag 3), consistent with a countywide policy shock, not LIHEAP, driving those values. These results are preliminary: each subset covers only 3–4 months, and a longer window would be needed to confirm them.